

# Longest Common Prefix

Input file:            **standard input**  
Output file:        **standard output**  
Time limit:        1 second  
Memory limit:     1024 megabytes

You are given  $n$  lowercase English strings  $s_1, s_2, \dots, s_n$ .

For each pair of integers  $(i, j)$  satisfying  $1 \leq j \leq i \leq n$ , define  $f_{i,j}$  as the maximum possible length of the longest common prefix of exactly  $j$  strings selected from the first  $i$  strings. Formally,

$$f_{i,j} = \max_{T \subseteq \{1, 2, \dots, i\}, |T|=j} |\text{LCP}(s_k \mid k \in T)|.$$

The longest common prefix of a single string is the string itself.

For each  $i = 1, 2, \dots, n$ , calculate

$$\sum_{j=1}^i (f_{i,j} \oplus j),$$

where  $\oplus$  denotes the bitwise XOR operation.

## Input

The first line contains an integer  $n$  ( $1 \leq n \leq 5 \cdot 10^5$ ), the number of strings.

Each of the next  $n$  lines contains a non-empty string  $s_i$  consisting only of lowercase English letters.

It is guaranteed that

$$\sum_{i=1}^n |s_i| \leq 5 \cdot 10^5.$$

## Output

Print  $n$  lines.

The  $i$ -th line should contain one integer:

$$\sum_{j=1}^i (f_{i,j} \oplus j).$$

## Examples

| standard input                                  | standard output                           |
|-------------------------------------------------|-------------------------------------------|
| 4<br>a<br>ab<br>abc<br>abcd                     | 0<br>6<br>4<br>12                         |
| 8<br>ab<br>b<br>ba<br>ab<br>aa<br>ba<br>a<br>bb | 3<br>5<br>9<br>10<br>14<br>20<br>28<br>36 |

## Note

The longest common prefix of several strings is the longest string that is a prefix of every selected string.

The symbol  $\oplus$  denotes the bitwise XOR operation.